

Background

- Smoking is associated with poor health outcomes and reduced work productivity.
- Little is known about effect of smoking history on labor force participation and absenteeism among working-age (i.e., 18-64 years) cancer survivors.

Methods

- **Data and sample:**
 - ❑ Pooled National Health Interview Survey 2013-2019, 2022-2023
 - ❑ Adults aged 18-64 years with a cancer history (n=8,203, weighted n=7,048,624)
- **Measures:**
 - ❑ Exposure: self-reported smoking history categorized as current, former, or never smoker
 - ❑ Outcomes: productivity loss measured as labor force participation, inability to work due to health, any missed workdays, number of missed workdays in the past 12 months
 - ❑ Key covariates: age, sex, marital status, education, body mass index (BMI), health insurance, time since diagnosis, survey year, and region
- **Analysis:**
 - ❑ Descriptive characteristics by smoking status (current, former, never)
 - ❑ Separate multivariable logistic (labor force participation, inability to work, and any missed workdays) and negative binomial (number of missed workdays) regressions examined associations between smoking status and productivity loss, adjusting for key covariates (see table)

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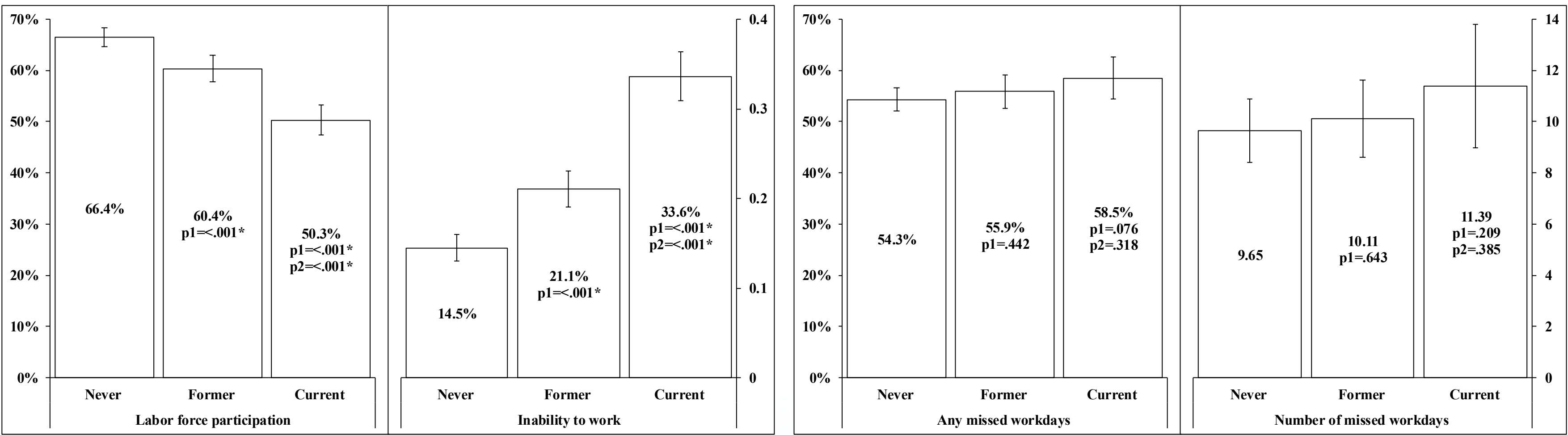
Characteristics of cancer survivors ages 18-64 years by smoking status

Individual-level characteristics (total n=8,203)	Currently smoke (n=1,793)		Formerly smoked (n=2,189)		Never smoked (reference group; n=4,221)
	%	P ^a	%	P ^a	%
Age (years)					
18-39	19.9	0.035	7.9	<.001	17.2
40-54	20.9		15.7		19.2
55-64	59.2		76.5		63.7
Sex					
Male	32.5	0.517	41.3	<.001	33.7
Female	67.5		58.7		66.3
Educational attainment					
< High school	18.2	<.001	8.8	0.003	7.5
High school graduate	34.4		23.3		19.5
≥ Some college	47.4		67.9		73.1
Marital status					
Married	55.7	<.001	69.6	0.870	69.9
Not married or others	44.3		30.4		30.1
Number of cancer diagnoses					
1 cancer diagnosis only	9.5	0.069	10.9	0.001	7.8
2-3 cancer diagnoses	90.5		89.1		92.2
BMI (kg/m ²)					
Normal weight (18.5-24.9)	33.7	<.001	27.4	0.700	28.0
Underweight (<18.5)	4.0		1.6		1.1
Overweight (25-29.)	31.8		32.6		33.1
Obese (≥30)	30.5		38.5		37.8
Recent Diagnosis (<2 years)					
Yes	15.9	0.653	17.8	0.328	16.5
No	84.1		82.2		83.5
Health insurance¶					
18-64, any private	42.8	<.001	70.0	<.001	76.2
18-64, public only	41.4		24.6		18.2
18-64, uninsured	15.8		5.4		5.6

Note: All statistical tests were two-sided and all p values were calculated using Pearson's chi-squared test statistic. US census region was included but not shown here.

Results

Crude prevalence productivity loss by smoking status among working-age cancer survivors



Adjusted productivity loss by smoking status among working-age cancer survivors

All cancer survivors (n=8,203)						
Smoking status	Labor force participation			Inability to work due to health		
	Adjusted % (95CI)	p-value compared to never smoked	p-value compared to formerly smoked	Adjusted mean (95CI)	p-value compared to never smoked	p-value compared to formerly smoked
Never	62.6% (60.9%, 64.3%)	reference	NA	18.0% (16.5%, 19.5%)	reference	NA
Former	61.4% (59.1%, 63.7%)	0.445	reference	20.0% (18.3%, 21.8%)	0.070	reference
Current	59.4% (56.7%, 62.2%)	0.061	0.120	23.9% (21.9%, 26.0%)	<0.001	0.001
Cancer survivors who were working last week (n=5,320)						
Smoking status	Any missed workdays in past 12 months			Number of missed workdays in past 12 months		
	Adjusted % (95CI)	p-value compared to never smoked	p-value compared to formerly smoked	Adjusted mean (95CI)	p-value compared to never smoked	p-value compared to formerly smoked
Never	53.8% (51.6%, 55.9%)	reference	NA	9.87 (95% CI: 8.61, 11.12)	reference	NA
Former	56.4% (53.2%, 59.5%)	0.180	reference	9.88 (95% CI: 8.47, 11.29)	0.987	reference
Current	59.7% (55.6%, 63.8%)	0.016	0.522	13.09 (95% CI: 9.63, 16.56)	0.092	0.043

Note: All p values were obtained from multivariable regressions.

Conclusions

- Working-age cancer survivors who currently smoke experience greater productivity loss compared with those who formerly and never smoked. Moreover, quitting smoking helps reduce the likelihood of experiencing inability to work and improve work absenteeism among working age cancer survivors.
- Findings have implications for improving smoking cessation efforts which may improve health and reduce productivity loss among working age cancer survivors.